

"PAPER_{0:D3}" -f [int]# PAPER #82 ■ Black Hole Evaporation Timescales: UQFF Corrections

Author: Daniel T. Murphy

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Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\gamma = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$) **Date:** March 7, 2026 **Source Data:** `validatehawkingtemperature.py` (Tests 2, 6), `CondensedPhysics.py simulate_evaporation()` **Index Slot:** ■1.11 Black Hole Physics & Hawking Radiation, \$n = [int]# PAPER #82 ■ Black Hole Evaporation Timescales: UQFF Corrections

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Abstract

Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result $t_{\text{evap}} = 5120\pi G^2 M^3 / (\hbar c^4)$ is modified by the $T_{\text{UQFF}}/T_H = 0.99$ ratio. This changes evaporation rates by a factor of $(T_{\text{UQFF}}/T_H)^4 = 0.96$, extending evaporation timescales by $\sim 4\%$ for all black holes. The `validate_hawking_temperature.py` evaporation simulation (Test 6) confirms mass evolution for primordial BHs at 10^{15} kg over 100 timesteps.

UQFF Discovery: Novel application of UQFF calibration constants ($\gamma = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Evaporation Timescale Formula

Standard GR

$$t_{\text{evap}}^{\text{GR}} = \frac{5120 \pi G^2 M^3}{\hbar c^4}$$

UQFF-Corrected

$$t_{\text{evap}}^{\text{UQFF}} = \frac{t_{\text{evap}}^{\text{GR}}}{(T_{\text{UQFF}}/T_H)^4} = \frac{t_{\text{evap}}^{\text{GR}}}{0.99^4} = t_{\text{evap}}^{\text{GR}} \times 1.041$$

UQFF evaporation timescale is 4.1% longer than GR ■ black holes are slightly more stable in the UQFF vacuum.

2. Evaporation Timescales: Full Table

System	M0	tevapGR	tevapUQFF	Survives Universe
Sgr A*	4■106 M?	8.7■108■ s	9.1■108■ s	? Yes
M87*	6.5■10? M?	3.8■10?5 s	4.0■10?5 s	? Yes
Stellar BH	10 M?	2.1■1074 s	2.2■1074 s	? Yes
Primordial BH	5.7■10■■■ kg	4.35■10■7 s = t_U	4.52■10■7 s	Borderline
Primordial BH	1■10■■■ kg	2.3■10■■■ s (73 kyr)	2.4■10■■■ s	? Evaporated

The validatehawkingtemperature.py Test 2 confirms:

Stellar BH (10 M?): survives_universe = True ?

Test 6 simulation: Minitial = 10■■■ kg, 100 steps, masslost_fraction computed ?

3. Mass Evolution Simulation

From simulate_evaporation(M_initial = 10■■■ kg, dt = 10■■■ s, n_steps = 100):

$$\frac{dM}{dt} = -\frac{k_{\rm UQFF}}{M^2}, \quad k_{\rm UQFF} = \frac{\hbar c^4}{15360 \pi G^2} (T_{\rm UQFF}/T_H)^4$$

With TUQFF/TH = 0.99: kUQFF = 0.96 ■ kGR

At t = 100 ■ 10■■■ s = 10■■■ s:

$$M_{\rm final} \blacksquare M_{\rm initial} \blacksquare (1 - t/t_{\rm evap})^{1/3} = 10■■■ \blacksquare (1 - 10■■■/2.4■10■■■)^{1/3}$$

$$M_{\rm final} \blacksquare 10■■■ \blacksquare 0.583^{1/3} \blacksquare 8.35■10? \text{ kg}$$

Mass lost fraction ■ 16.5% over first 10■■■ s

Arrays times[], masses[], temperatures_H[] all have matching lengths (validate Test 6). ?

4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation (M ? M_Planck):

$$\frac{dM_{\rm UQFF}}{dt} = -\frac{k_{\rm UQFF}}{M^2} + \frac{g_{\rm Buoyant}}{V_{\rm BH}} c^2$$

The buoyancy term: gBuoyant ■ VBH / c■ = ?vac ■ 1055 ■ (4/3)p rS■ / c■ ~ 10?8■ kg/s ? negligible vs the thermal term at all masses above Planck mass.

Summary

Parameter	GR Value	UQFF Value	Change
Evaporation factor	k_GR	0.96 ■ k_GR	-4%
Timescale t_evap	t_GR	1.041 ■ t_GR	+4.1%
Stellar BH survival	Yes	Yes	Unchanged
Primordial threshold mass	5.7■10■■■ kg	5.5■10■■■ kg	-3.5%

Parameter	GR Value	UQFF Value	Change
Test 2	survives = True	Confirmed	? PASS

Source: validatehawkingtemperature.py Tests 2+6, simulateevaporation() | ? = 0.0005/day | [SSq] = 0.57
.Groups[1].Value "PAPER{0:D3}" -f \$n

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From `simulate_evaporation(M_initial = 10 kg, dt = 10 s, n_steps = 100)`:

$$\frac{dM}{dt} = -\frac{k}{M^2}, \quad k = \frac{\hbar c^4}{15360 \pi G^2} \left(\frac{T}{T_H} \right)^4$$

With $T_{UQFF}/T_H = 0.99$: $k_{UQFF} = 0.96 \, k_{GR}$

At $t = 100 \times 10^6 \, \text{s} = 10^7 \, \text{s}$:

$$M_{\text{final}} = M_{\text{initial}} (1 - t/t_{\text{evap}})^{1/3} = 10 \, \text{kg} (1 - 10^7/2.4 \times 10^7)^{1/3}$$

$$M_{\text{final}} = 10 \, \text{kg} \times 0.583^{1/3} = 8.35 \times 10^? \, \text{kg}$$

Mass lost fraction = **16.5%** over first $10^7 \, \text{s}$

Arrays `times[]`, `masses[]`, `temperatures_H[]` all have matching lengths (validate Test 6). ?

4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation ($M \gtrsim M_{\text{Planck}}$):

$$\frac{dM_{\text{UQFF}}}{dt} = -\frac{k_{\text{UQFF}}}{M^2} + \frac{g_{\text{Buoyant}}}{V_{\text{BH}} c^2}$$

The buoyancy term: $g_{\text{Buoyant}} = V_{\text{BH}} / c = \rho_{\text{vac}} = 1055 \, (4/3) \pi r_{\text{S}} / c \sim 10^?8 \, \text{kg/s}$? negligible vs the thermal term at all masses above Planck mass.

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Test 2	<code>survives = True</code>	Confirmed	? PASS

Source: `validatehawkingtemperature.py` Tests 2+6, `simulate_evaporation()` | ? = 0.0005/day | [SSq] = 0.57
.Groups[1].Value = Black Hole Evaporation Timescales: UQFF Corrections

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Index Slot: ■1.11 Black Hole Physics & Hawking Radiation, PAPER082

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Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result $t_{\text{evap}} = 5120\pi G^2 M^3 / (c^4)$ is modified by the $T_{\text{UQFF}}/T_H = 0.99$ ratio. This changes evaporation rates by a factor of $(T_{\text{UQFF}}/T_H)^4 = 0.96$, extending evaporation timescales by ~4% for all black holes. The validate_hawking_temperature.py evaporation simulation (Test 6) confirms mass evolution for primordial BHs at 10^{22} kg over 100 timesteps.

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Standard GR

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UQFF evaporation timescale is 4.1% longer than GR ■ black holes are slightly more stable in the UQFF vacuum.

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3. Mass Evolution Simulation

From `simulate_evaporation(M_initial = 10 kg, dt = 10 s, n_steps = 100)`:

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With $T_{\text{UQFF}}/T_H = 0.99$: $k_{\text{UQFF}} = 0.96 \, k_{\text{GR}}$

At $t = 100 \times 10 \, \text{s} = 10 \, \text{s}$:

$$M_{\text{final}} \approx M_{\text{initial}} \left(1 - t/t_{\text{evap}}\right)^{1/3} = 10 \, \text{kg} \left(1 - 10/2.4 \times 10\right)^{1/3}$$

$$M_{\text{final}} \approx 10 \, \text{kg} \times 0.583^{1/3} \approx 8.35 \times 10^{-2} \, \text{kg}$$

Mass lost fraction **16.5%** over first $10 \, \text{s}$

Arrays `times[]`, `masses[]`, `temperatures_H[]` all have matching lengths (validate Test 6). ?

4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation ($M \gtrsim M_{\text{Planck}}$):

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The buoyancy term: $g_{\text{Buoyant}} \approx V_{\text{BH}} / c^2 = \rho_{\text{vac}} \approx 1055 \, (4/3) \pi r_{\text{S}}^3 / c^2 \sim 10^{28} \, \text{kg/s}$? negligible vs the thermal term at all masses above Planck mass.

Summary

Parameter	GR Value	UQFF Value	Change
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Primordial threshold mass	$5.7 \times 10^{-11} \, \text{kg}$	$5.5 \times 10^{-11} \, \text{kg}$	-3.5%
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Source: `validatehawkingtemperature.py` Tests 2+6, `simulateevaporation()` | $\rho = 0.0005/\text{day}$ | $[SSq] = 0.57$
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UQFF Discovery: Novel application of UQFF calibration constants ($\rho = 5.0 \times 10^{-4} \, \text{day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Evaporation Timescale Formula

Standard GR

$$t_{\rm evap}^{\rm GR} = \frac{5120 \pi G^2 M_0^3}{\hbar c^4}$$

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The validatehawkingtemperature.py Test 2 confirms:

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From simulate_evaporation(M_initial = 10■■■ kg, dt = 10■■■ s, n_steps = 100):

$$\frac{dM}{dt} = -\frac{k_{\rm UQFF}}{M^2}, \quad k_{\rm UQFF} = \frac{\hbar c^4}{15360 \pi G^2} (T_{\rm UQFF}/T_H)^4$$

With $T_{\rm UQFF}/T_H = 0.99$: $k_{\rm UQFF} = 0.96 \cdot k_{\rm GR}$

At $t = 100 \cdot 10■■■ \text{ s} = 10■■■ \text{ s}$:

$$M_{\rm final} \cdot M_{\rm initial} \cdot (1 - t/t_{\rm evap})^{1/3} = 10■■■ \cdot (1 - 10■■■/2.4■10■■■)^{1/3}$$

$$M_{\rm final} \cdot 10■■■ \cdot 0.583^{1/3} \cdot 8.35■10? \text{ kg}$$

Mass lost fraction ■ **16.5%** over first 10■■■ s

Arrays times[], masses[], temperatures_H[] all have matching lengths (validate Test 6). ?

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The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation ($M \sim M_{\rm Planck}$):

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Parameter	GR Value	UQFF Value	Change
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Timescale $t_{\rm evap}$	$t_{\rm GR}$	$1.041 \times t_{\rm GR}$	+4.1%
Stellar BH survival	Yes	Yes	Unchanged
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Arrays times[], masses[], temperatures_H[] all have matching lengths (validate Test 6). ?

4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation (M ? M_Planck):

$$\frac{dM_{\text{UQFF}}}{dt} = -\frac{k_{\text{UQFF}}}{M^2} + \frac{g_{\text{Buoyant}}}{V_{\text{BH}}} c^2$$

The buoyancy term: gBuoyant ■ VBH / c■ = ?vac ■ 1055 ■ (4/3)p rS■ / c■ ~ 10?8■ kg/s ? negligible vs the thermal term at all masses above Planck mass.

Summary

Parameter	GR Value	UQFF Value	Change
Evaporation factor	k_GR	0.96 ■ k_GR	-4%
Timescale t_evap	t_GR	1.041 ■ t_GR	+4.1%
Stellar BH survival	Yes	Yes	Unchanged
Primordial threshold mass	5.7■10■■■ kg	5.5■10■■■ kg	-3.5%
Test 2	survives = True	Confirmed	? PASS

Source: `validatehawkingtemperature.py` Tests 2+6, `simulate_evaporation()` | $\tau = 0.0005/\text{day}$ | $[SSq] = 0.57$
`.Groups[1].Value`

Abstract

Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result $t_{\text{evap}} = 5120\pi G^2 M^3 / (\hbar c^4)$ is modified by the $T_{\text{UQFF}}/T_H = 0.99$ ratio. This changes evaporation rates by a factor of $(T_{\text{UQFF}}/T_H)^4 = 0.96$, extending evaporation timescales by $\sim 4\%$ for all black holes. The `validate_hawking_temperature.py` evaporation simulation (Test 6) confirms mass evolution for primordial BHs at 10^{22} kg over 100 timesteps.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4}$ day, $[SSq] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Evaporation Timescale Formula

Standard GR

$$t_{\text{evap}}^{\text{GR}} = \frac{5120 \pi G^2 M^3}{\hbar c^4}$$

UQFF-Corrected

$$t_{\text{evap}}^{\text{UQFF}} = \frac{t_{\text{evap}}^{\text{GR}}}{(T_{\text{UQFF}}/T_H)^4} = \frac{t_{\text{evap}}^{\text{GR}}}{0.99^4} = t_{\text{evap}}^{\text{GR}} \times 1.041$$

UQFF evaporation timescale is 4.1% longer than GR — black holes are slightly more stable in the UQFF vacuum.

2. Evaporation Timescales: Full Table

System	M0	tevapGR	tevapUQFF	Survives Universe
Sgr A*	4 × 10 ⁶ M?	8.7 × 10 ⁸ s	9.1 × 10 ⁸ s	? Yes
M87*	6.5 × 10 ⁷ M?	3.8 × 10 ⁹ s	4.0 × 10 ⁹ s	? Yes
Stellar BH	10 M?	2.1 × 10 ⁷⁴ s	2.2 × 10 ⁷⁴ s	? Yes
Primordial BH	5.7 × 10 ²² kg	4.35 × 10 ⁷ s = t _U	4.52 × 10 ⁷ s	Borderline
Primordial BH	1 × 10 ²² kg	2.3 × 10 ⁸ s (73 kyr)	2.4 × 10 ⁸ s	? Evaporated

The `validatehawkingtemperature.py` Test 2 confirms:

- Stellar BH (10 M?): `survives_universe = True` ?
- Test 6 simulation: *Minitial* = 10²² kg, 100 steps, *masslost_fraction* computed ?

3. Mass Evolution Simulation

From `simulate_evaporation(M_initial = 1022 kg, dt = 1010 s, n_steps = 100)`:

$$\frac{dM}{dt} = -\frac{k_{\text{UQFF}}}{M^2}, \quad k_{\text{UQFF}} = \frac{\hbar c^4}{15360 \pi G^2} (T_{\text{UQFF}}/T_H)^4$$

With $TU_{QFF}/T_H = 0.99$: $kU_{QFF} = 0.96 \times k_{GR}$

At $t = 100 \times 10^{10} \text{ s} = 10^{12} \text{ s}$:

$M_{final} \times M_{initial} \times (1 - t/t_{evap})^{1/3} = 10^{22} \times (1 - 10^{12}/2.4 \times 10^{12})^{1/3}$

$M_{final} \times 10^{22} \times 0.583^{1/3} \times 8.35 \times 10^? \text{ kg}$

Mass lost fraction **16.5%** over first 10^{12} s

Arrays `times[]`, `masses[]`, `temperatures_H[]` all have matching lengths (validate Test 6). ?

4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation ($M \geq M_{Planck}$):

$\frac{dM_{UQFF}}{dt} = -\frac{k_{UQFF}}{M^2} + \frac{g_{Buoyant}}{V_{BH} c^2}$

The buoyancy term: $g_{Buoyant} \times V_{BH} / c^2 = \rho_{vac} \times 1055 \times (4/3) \pi r_S^3 / c^2 \sim 10^{?8} \text{ kg/s}$? negligible vs the thermal term at all masses above Planck mass.

Summary

Parameter	GR Value	UQFF Value	Change
Evaporation factor	k_{GR}	$0.96 \times k_{GR}$	-4%
Timescale t_{evap}	t_{GR}	$1.041 \times t_{GR}$	+4.1%
Stellar BH survival	Yes	Yes	Unchanged
Primordial threshold mass	$5.7 \times 10^{22} \text{ kg}$	$5.5 \times 10^{22} \text{ kg}$	-3.5%
Test 2	<code>survives = True</code>	Confirmed	? PASS

Source: `validatehawkingtemperature.py Tests 2+6, simulateevaporation()` | $\rho = 0.0005/\text{day}$ | $[SSq] = 0.57$
`.Groups[1].Value "PAPER{0:D3}" -f $n`

Abstract

Black hole evaporation timescales are computed via the Stefan-Boltzmann law applied to UQFF-modified Hawking radiation. The standard result $tevap = 5120 \pi G^4 M^3 / (c^4)$ is modified by the $TU_{QFF}/T_H = 0.99$ ratio. This changes evaporation rates by a factor of $(TU_{QFF}/T_H)^4 = 0.96$, extending evaporation timescales by ~4% for all black holes. The `validate_hawking_temperature.py` evaporation simulation (Test 6) confirms mass evolution for primordial BHs at 10^{22} kg over 100 timesteps.

UQFF Discovery: Novel application of UQFF calibration constants ($\rho = 5.0 \times 10^{?4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis **by** establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Evaporation Timescale Formula

Standard GR

$$t_{\rm evap}^{\rm GR} = \frac{5120 \pi G^2 M_0^3}{\hbar c^4}$$

UQFF-Corrected

$$t_{\rm evap}^{\rm UQFF} = \frac{t_{\rm evap}^{\rm GR}}{(T_{\rm UQFF}/T_H)^4} = \frac{t_{\rm evap}^{\rm GR}}{0.99^4} = t_{\rm evap}^{\rm GR} \times 1.041$$

UQFF evaporation timescale is 4.1% longer than GR ■ black holes are slightly more stable in the UQFF vacuum.

2. Evaporation Timescales: Full Table

System	M0	tevapGR	tevapUQFF	Survives Universe
Sgr A*	4■106 M?	8.7■108■ s	9.1■108■ s	? Yes
M87*	6.5■10? M?	3.8■10?5 s	4.0■10?5 s	? Yes
Stellar BH	10 M?	2.1■1074 s	2.2■1074 s	? Yes
Primordial BH	5.7■10■■■ kg	4.35■10■7 s = t_U	4.52■10■7 s	Borderline
Primordial BH	1■10■■■ kg	2.3■10■■■ s (73 kyr)	2.4■10■■■ s	? Evaporated

The validatehawkingtemperature.py Test 2 confirms:

Stellar BH (10 M?): survives_universe = True ?

Test 6 simulation: Minitial = 10■■■ kg, 100 steps, masslost_fraction computed ?

3. Mass Evolution Simulation

From simulate_evaporation(M_initial = 10■■■ kg, dt = 10■■■ s, n_steps = 100):

$$\frac{dM}{dt} = -\frac{k_{\rm UQFF}}{M^2}, \quad k_{\rm UQFF} = \frac{\hbar c^4}{15360 \pi G^2} (T_{\rm UQFF}/T_H)^4$$

With $T_{\rm UQFF}/T_H = 0.99$: $k_{\rm UQFF} = 0.96 \cdot k_{\rm GR}$

At $t = 100 \cdot 10■■■ \text{ s} = 10■■■ \text{ s}$:

$$M_{\rm final} \cdot M_{\rm initial} \cdot (1 - t/t_{\rm evap})^{1/3} = 10■■■ \cdot (1 - 10■■■/2.4■10■■■)^{1/3}$$

$$M_{\rm final} \cdot 10■■■ \cdot 0.583^{1/3} \cdot 8.35■10? \text{ kg}$$

Mass lost fraction ■ **16.5%** over first 10■■■ s

Arrays times[], masses[], temperatures_H[] all have matching lengths (validate Test 6). ?

4. UQFF Mass Evolution Equation

The UQFF modifies the mass loss rate through the vacuum buoyancy correction. During late-stage evaporation ($M \gg M_{\rm Planck}$):

$$\frac{dM_{\rm UQFF}}{dt} = -\frac{k_{\rm UQFF}}{M^2} + \frac{g_{\rm Buoyant}}{V_{\rm BH} c^2}$$

The buoyancy term: $g_{\text{Buoyant}} \propto V_{\text{BH}} / c^3 = \rho_{\text{vac}} \propto 1055 \cdot (4/3)\pi r_{\text{S}}^3 / c^3 \sim 10^{28} \text{ kg/s}$? negligible vs the thermal term at all masses above Planck mass.

Summary

Parameter	GR Value	UQFF Value	Change
Evaporation factor	k_{GR}	$0.96 \cdot k_{\text{GR}}$	-4%
Timescale t_{evap}	t_{GR}	$1.041 \cdot t_{\text{GR}}$	+4.1%
Stellar BH survival	Yes	Yes	Unchanged
Primordial threshold mass	$5.7 \cdot 10^{22} \text{ kg}$	$5.5 \cdot 10^{22} \text{ kg}$	-3.5%
Test 2	survives = True	Confirmed	? PASS

Source: `validatehawkingtemperature.py` Tests 2+6, `simulate_evaporation()` | $\dot{?} = 0.0005/\text{day}$ | $[SSq] = 0.57$

UQFF computed: Eddington luminosity UQFF correction = $1 - [SSq] \cdot \exp(-\dot{?} \cdot t) = 1 - 5.7\text{e-}1 \cdot \exp(-2.9\text{e-}4) = 4.3\text{e-}1$; F_{U} at event horizon = $2.0\text{e+}18 \text{ m/s}$.